Deepika Pingali

Ph.D. Candidate, Agricultural and Applied Economics
University of Illinois Urbana–Champaign

Contact

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Education

Ph.D. in Agricultural and Applied Economics (ABD) 2022 – Present
Department of Agricultural and Consumer Economics (ACE),
University of Illinois Urbana–Champaign
GPA: 4.0/4.0
Graduate Minor: Gender Relations in International Development (GRID)

Master of Business Administration 2014 – 2016
Indian Institute of Management (IIM) Ahmedabad, India

B.E. (Honors), Electronics & Communication Engineering 2010 – 2014
Birla Institute of Technology and Science (BITS) Pilani, India

Research Fields

Development Economics • Data Science • Political Economy • Economics of Networks

Technical Skills

Quantitative Skills: Data Science, Machine Learning, Causal Inference, Geospatial Analysis, Social Network Analysis, Network Econometrics
Software: Stata, R, Python, MATLAB

Publications

- Lyons, A., Kass-Hanna, J., **Pingali, D.**, Soliman, A., Zhu, D., Zhang, Y., & Montoya Castano, A. (2025). A geospatial analysis of food insecurity among refugee households in Lebanon using machine learning techniques. *Food Security.* (Earlier version circulated as ERF Working Paper No. 1729, Economic Research Forum.)
- Verma, P., Dengler, N., Lyons, A., & **Pingali, D.**(forthcoming). Beyond gravity models: Predicting human migration with spatial gates and graph neural networks. In *Proceedings of the 2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL '26).*

Working Papers

- **Pingali, D.**(2026). *Uncovering the influence of hidden social networks on gendered labor norms: Evidence from rural India* [Job Market Paper].

  Abstract: Economics has traditionally characterized spousal labor sharing as the outcome of an intrahousehold bargaining process; however, these decisions are also shaped by norms embedded within and transmitted through social networks. This paper studies the influence of peer behavior on spouses' relative contributions to domestic and production labor in rural India. A key empirical challenge is that social network ties are unobserved in most existing datasets. I overcome this challenge by jointly estimating the latent social structure, peer effects, and contextual effects within a linear social interactions model. I use an Adaptive Elastic Net Generalized Method of Moments, recovering network architecture without requiring directly observed network data. Using the Village Dynamics in South Asia (VDSA) dataset across 16 villages and a balanced panel of 629 households, I find that within-sample peer networks governing production and domestic labor-sharing differ in topology. Both, domestic labor sharing and production labor sharing exhibit a statistically significant social multiplier. A 10% increase in peers' relative male-to-female domestic work leads to a 6.2% increase in a couple's own relative male-to-female domestic work. Similarly, a 10% increase in peers' relative female-to-male production work leads to a 9.3% increase in a couple's own relative female-to-male production work. Exploratory analysis of the recovered networks shows that couples maintain influence ties with roughly five peers on average, and that this degree is stable across village sizes. I also examine assortativity, the tendency to connect with similar peers. Domestic-norm networks are positively religion- and land-assortative, meaning same-religion and similarly landed couples are tied more often than chance would predict; production-norm networks show no such assortativity on either dimension. Women's participation in self-help groups is positively associated with centrality in production-norm networks, whereas landlessness is negatively associated with it. In contrast, higher centrality in domestic-norm networks is associated with a larger spousal age gap, greater female education, and lower-to-middle landholding status. My findings provide novel insights and evidence on how network-mediated spillover effects are integral in shaping intrahousehold labor distributions.

- **Pingali, D.**(2025). *The effect of political capital on migration decisions in the context of climate change in India* [Working paper]. Available at SSRN: https://ssrn.com/abstract=5500739

  Abstract: Climate change is increasingly reshaping livelihoods through droughts, crop failures, rising sea levels, and extreme weather events, particularly in South Asia, Sub-Saharan Africa, and Latin America. Migration has become a crucial household strategy for managing these climate-induced risks. This paper examines how village-level political capital — households' close connections to local government officials — shapes migration decisions under environmental stress, and how this influence varies with the intensity of long-term climate stress. We develop a stylized expected-utility framework in which households weigh expected gains from migration against the costs and risks of remaining in place. The framework predicts a nonlinear, inverted U-shaped relationship: political capital has the largest effect at moderate levels of climate stress. Using data from the India Human Development Survey and ERA5 climate records, we implement a difference-in-differences approach. Empirical results confirm the model: at medium heat stress, political capital increases migration probability by 3.2 percentage points ($p < 0.05$), while its effect is smaller and statistically insignificant under low or extreme stress. These findings demonstrate that political access mediates climate adaptation, with implications for

equitable migration policies and resilience strategies.

- Lyons, A., Polato e Fava, A., Kass-Hanna, J., **Pingali, D.**, & Liu, F. (2025). *Gender disparities in financial and digital financial inclusion: Insights from PISA data* [Working paper, under review]. Available at SSRN: https://ssrn.com/abstract=5384658

  Abstract: The global digital transformation of financial services has heightened concerns about gender disparities in financial inclusion and empowerment, particularly as gender gaps in financial literacy often emerge early in life. This study leverages data from the 2018 PISA assessment to examine gender differences in the relationship between financial literacy and both traditional and digital financial inclusion among youth. Using the Lewbel method to address endogeneity, the analysis reveals that financial literacy significantly enhances financial and digital financial inclusion. However, persistent gender disparities remain, especially in traditional financial inclusion, where girls are less likely than boys to own bank accounts or payment cards. Gaps in confidence are also notable, with girls up to 50 percentage points less likely to feel confident performing financial and digital financial tasks compared to boys. Encouragingly, participation in a financial literacy course significantly improves girls' confidence, highlighting its potential to reduce these disparities. Yet greater confidence does not necessarily translate to increased financial service participation, suggesting the continued presence of external barriers to financial inclusion. These findings underscore the need for targeted policy interventions to promote early financial education, digital skill development, and confidence-building among girls, laying a foundation for their economic empowerment in the digital age.

Works in Progress

- Lyons, A., **Pingali, D.**, & Soliman, A. *Using Machine Learning Techniques to Predict Internal Migration in the Brazilian Amazon.*
- Tripathy, P., Upadhyay, S., Alegbeleye, O. M., Senkardesler, E., **Pingali, D.**, Goddard, E., Thaker, J., Leiserowitz, A., & Marlon, J. *Mismatch Between Perceived and Assessed Flood Risk in India.*
- Lyons, A., **Pingali, D.**, Telles, T. S., Plassa, W., Soliman, A., & Storoe, B. *From Forest to Farm? Deforestation, Farm Lifecycles, and Long-Run Agricultural Sustainability in Brazil.*
- Lyons, A., **Pingali, D.**, Kass-Hanna, J., & Soliman, A. *Where Refugees Settle: Spatial Dependence and the Determinants of Syrian Refugee Distribution in Lebanon.*

Conference & Poster Presentations

- "Uncovering the Influence of Hidden Social Networks on Gendered Labor Norms: Evidence from Rural India" (sole author) [Job Market Paper].
  - ASSA/AEA Annual Meeting, Washington, DC, January 2027 (scheduled)
- "The Effect of Political Capital on Migration Decisions in the Context of Climate Change in India" (sole author).
  - Political Economy of Climate and Environment (PECE), Harvard University, Cambridge, MA, September 2026
  - Environmental and Climate Mobilities Network (ECMN), Wageningen University & Research, Wageningen, Netherlands, June 2026

  - European Political Science Society (EPSS), Belfast, United Kingdom, June 2026
  - Illinois Economics Association Conference, Chicago, IL, October 2025
- "Gender Disparities in Financial and Digital Inclusion: Insights from PISA Data."
  - Congreso Internacional en Alfabetización Financiera, Online, November 2025
  - International Conference on Empirical Economics, Penn State Altoona (Online), August 2025
  - Illinois Economics Association Conference, Chicago, IL, October 2024
- "A Geospatial Analysis of Food Insecurity Among Refugee Households in Lebanon Using Machine Learning Techniques."
  - ERF 30th Annual Conference, Rabat, Morocco, April 2024 (presented by A. Lyons)
  - Illinois Economics Association Conference, Chicago, IL, October 2024

Research and Teaching Experience

Research Assistant 2022 – Present
Department of Agricultural and Consumer Economics (ACE) & National Center for Supercomputing Applications (NCSA), University of Illinois Urbana–Champaign

- Leads research on forced displacement, climate change, and the socioeconomic vulnerability of disadvantaged populations, producing peer-reviewed publications, working papers, and presentations at international venues.
- Mentors and manages undergraduate researchers through NCSA's REU (Research Experiences for Undergraduates) and SPIN (Students Pushing INnovation) programs, as well as graduate students; mentees have gone on to co-author peer-reviewed publications and working papers arising from this research.

Teaching Assistant, ACE 474: Economics of Consumption Fall 2022
Department of Agricultural and Consumer Economics (ACE),
University of Illinois Urbana–Champaign

- Guided student teams through a Consumer Price Index (CPI) project modeled on U.S. Bureau of Labor Statistics methodology, advising them on constructing a custom price index to capture how college students' consumption baskets have shifted amid recent tariffs and rising inflation.
- Graded assignments and exams and held regular office hours to support student learning.

Teaching Assistant, ACE 270: Consumer Economics Spring 2024
Department of Agricultural and Consumer Economics (ACE),
University of Illinois Urbana–Champaign

- Managed grading responsibilities for assignments and exams.

Teaching Assistant, ACE 476: Behavioral Economics Fall 2026
Department of Agricultural and Consumer Economics (ACE),
University of Illinois Urbana–Champaign

- Support a semester-long student project by facilitating in-class activities held periodically to build student progress.
- Grade quizzes, homework, exams and activities and provide feedback to guide students' work on the project.

Fellowships & Awards

- Graduate Conference Travel Award, ACE — Summer 2026
- Wilken Farm Management Fellowship, ACE — Spring 2025
- University Block Grant Fellowship, ACE — Spring 2025
- Linse Scholarship, ACE — Spring 2024

Training and Workshops (Competitive)

- Winter School on Econometrics of Networks, VU Amsterdam, Online — 2025
- Oxford Summer School on Economic Networks, Oxford, UK — 2025
- I-GUIDE Summer School, Spatial AI for Extreme Events and Disaster Resilience, UCAR, Boulder, CO (NSF-supported) — 2025
- CSWEP South PhD Student Mentoring Workshop, Houston, TX (scheduled) — 2026

Service & Leadership

- Vice President, ACE Graduate Student Organization (GSO) — 2025 – 2026
- Secretary, ACE Graduate Student Organization (GSO) — 2023 – 2025

Other Professional Experience

Over five years in the international development sector across South Asia: managing grassroots programs and a social business, and advising INGOs and multilateral agencies. Areas of work include monitoring, evaluation and learning (MEL), program design and implementation, advisory services, fundraising, and branding & marketing.

Social Entrepreneurship

Founder, Vidhignya Enterprises Pvt. Ltd. — 2017 – 2022

- Founded a venture promoting collective entrepreneurship among women from the informal economy, empowering them through skill-building and livelihoods interventions and connecting them to sustainable market linkages.

Development Consulting — 2017 – 2022

- World Bank: land contracts and migration trends in India.
- World Bank & Access Development Services: supply-side financing barriers for women entrepreneurs.
- International Labour Organization: worker networks and labor conditions, New Delhi.
- UNDP: vision document for May Doe Kabar, a network of 20,000 rural women in Myanmar.
- DFID Global Innovation Fund: case studies of Indian firms.
- EU GENIE Program: review of the role of women's civil society organizations in India.
- Bangladesh-based INGO: design and review of a community empowerment program.
- Evaluation of international labor migration projects across Lebanon, Jordan, and South Asia.

Program Design & Implementation 2017 – 2022

- Directed grassroots anti-trafficking projects in Nepal (UN Trust Fund-funded NGO partnership), rescuing and rehabilitating trafficking survivors and supporting informal-sector workers through community networks.
- Directed a women's empowerment project in Varanasi and Koppal, India, benefiting 7,000 rural women through self-sustaining collectives.

Private Sector Experience

True North Private Equity, Mumbai 2016 – 2017

- Drove business development initiatives, launching and expanding new business verticals and shaping corporate strategy.

References

Available upon request.